

OpsBrain AI

Unified Asset & Operations Brain for Zero-Harm Plants

ET AI Hackathon 2026 | Problem Statement #8

Operational Silos & SCADA Blindspots

- **Sensor Blind Spots**

SCADA alarms warn of pressure spikes but do not know isolation valve status or the mandated safety steps.

- **The Knowledge Cliff**

Retiring senior engineers carry years of undocumented plant heuristics away, leaving junior operators with sparse guidance.

- **Audit Latency**

Cross-referencing safety compliance logs manually during an alert takes hours, slowing emergency response.

SCADA Telemetry

Out-of-context sensor readings

Regulatory SOPs

Trapped in static PDF binders

Asset Blueprints

Rasterized blueprints
disconnected from state

Incident Logs

Unstructured notebooks & logs

OpsBrain AI Architecture

- Physical Topology Mapping

Extracts connection rules and node properties directly from Piping & Instrumentation Diagrams (P&IDs).

- Local Vector Indexing

Ingests environmental and safety SOPs locally on CPU using BGE embeddings for absolute data privacy.

- Orchestration Engine

Deploys a fleet of LLM agents to monitor risks and offer prototype compliance evidence support.



Knowledge Graph Digital Twin

- **Gemini Vision Parser**

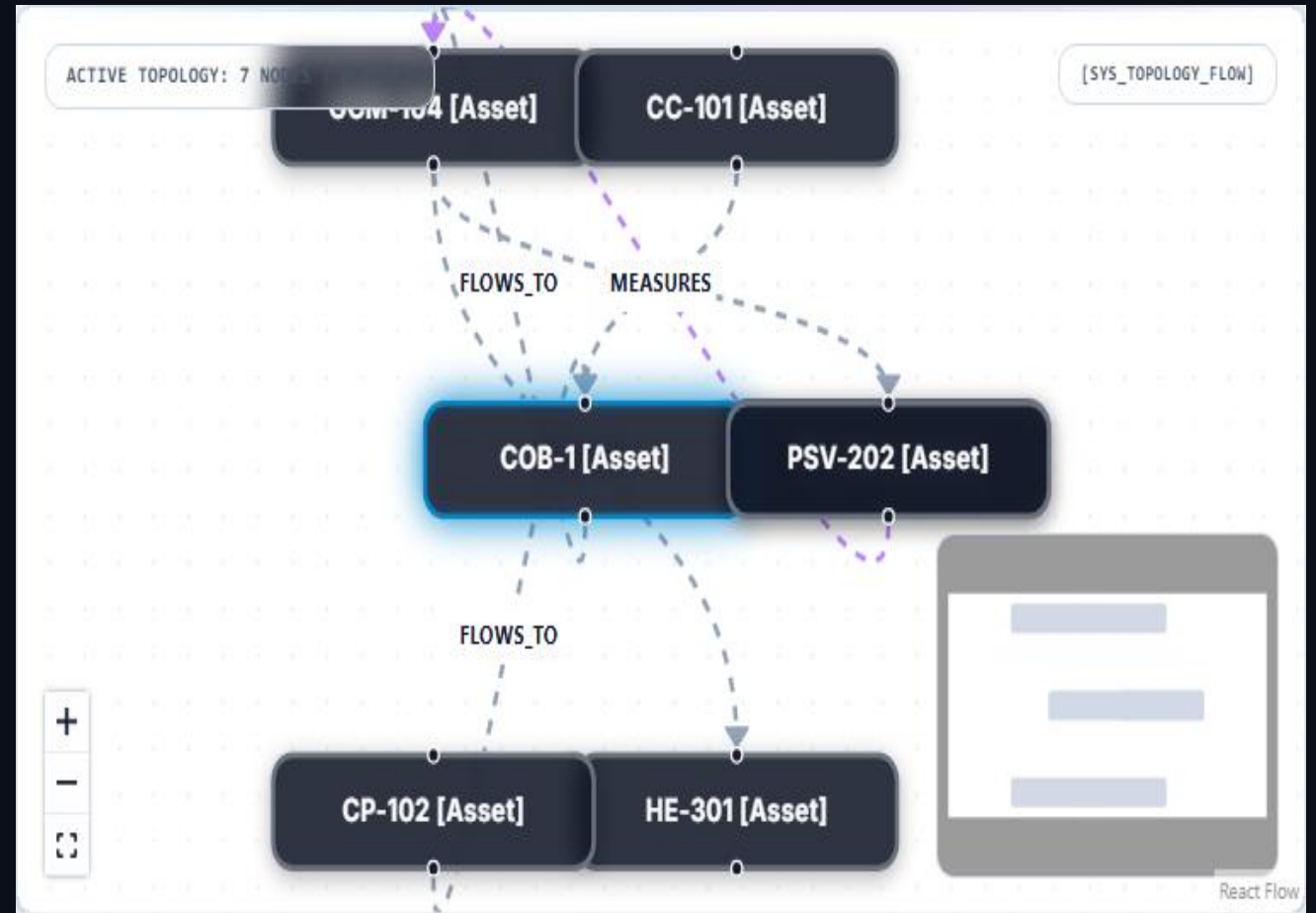
Vision model parses blueprint symbols, coordinates, flow directions, and equipment tags from drawings.

- **Relational Graph**

Renders equipment tags, flow lines, and sensor limits dynamically using ReactFlow.

- **Local Security**

Documents and asset links are stored on-premise, bypassing cloud risk vectors for heavy industries.



Specialized Multi-Agent Core

- Task-Focused Agents

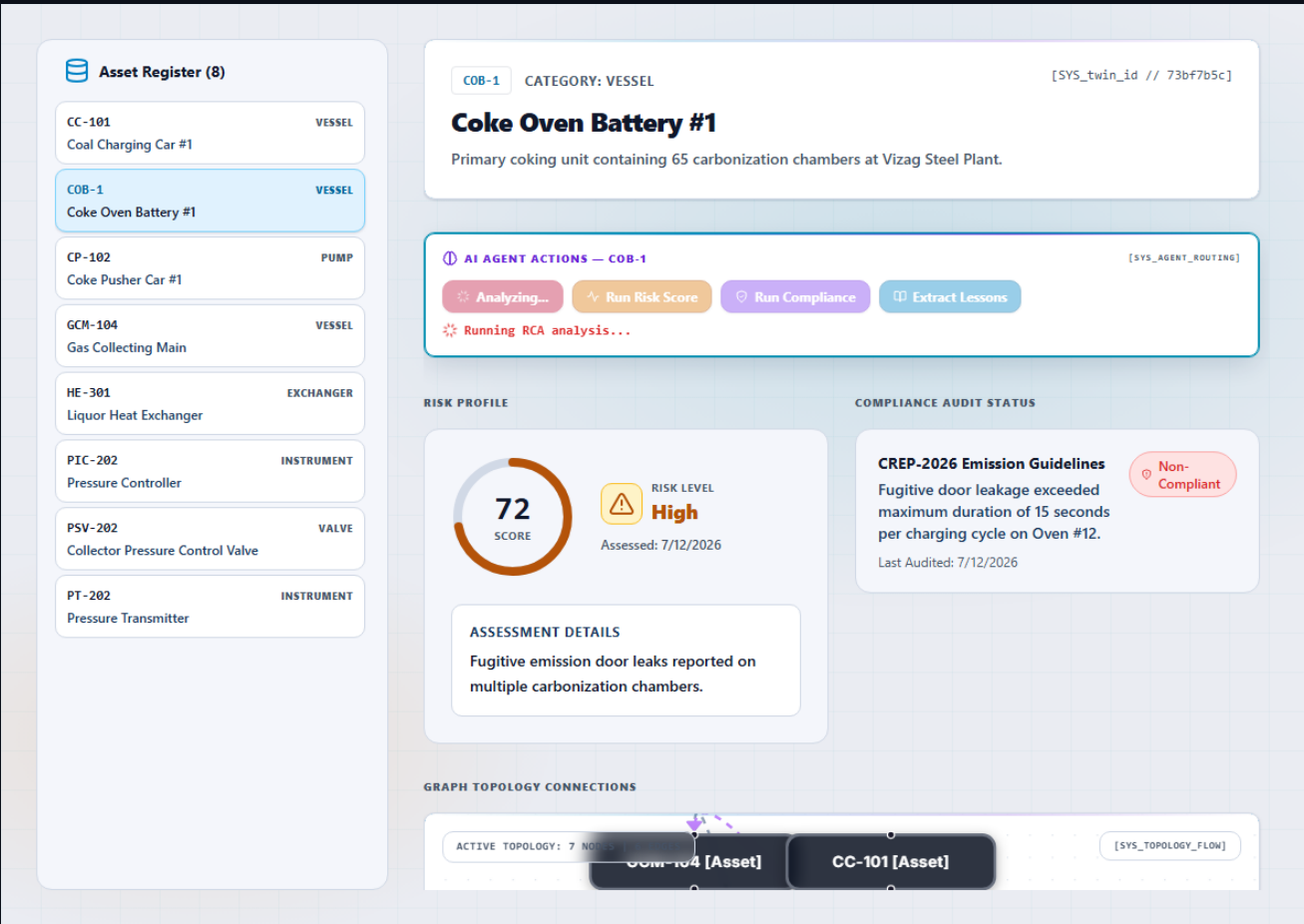
Five specialized agents (Root Cause, Risk, Compliance, Lessons, Copilot) handle targeted safety computations.

- AI Traversal HUD

Console overlay blocks screen during investigation to display detailed agent query and logic logs.

- Dynamic Graph Pulse

Pulsing warn highlights along traversed ReactFlow nodes track risk cascades on screen in real time.



Knowledge Copilot & Graph RAG

- **Graph-Aware Evidence Path**

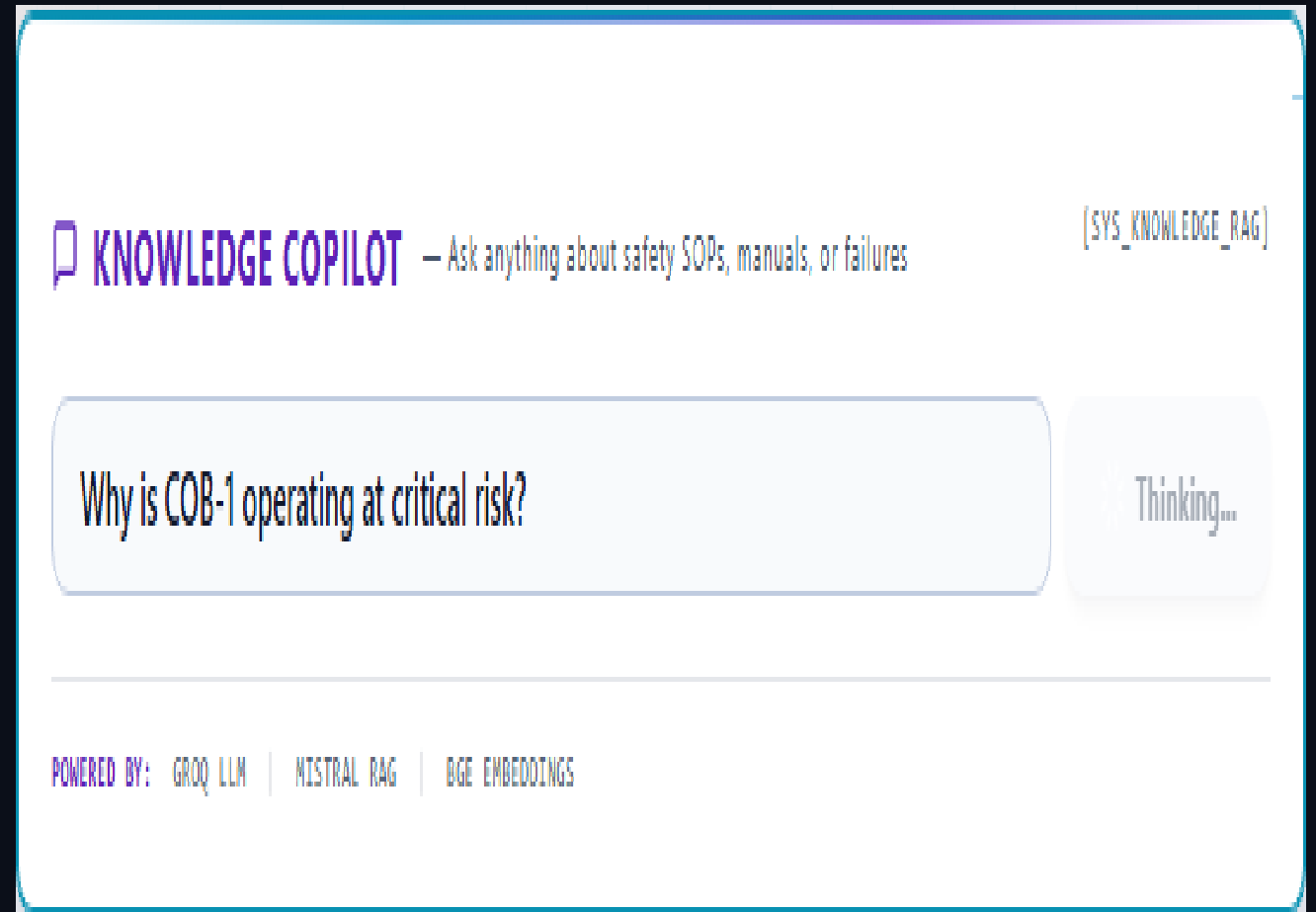
Maps queries onto physical connections, providing logical evidence paths across nodes.

- **Trust & Verifiability**

Returns direct SOP references, confidence percentages, and source document links.

- **Tribal Notes Panel**

Enables operators to type local notes on the twin, which are instantly indexed in the local vector store.



Provider Fallback & Runtime Monitor

- **Circuit-Breaker Logic**

Swaps completions (Groq -> Mistral -> Gemini) dynamically to ensure high availability.

- **Offline Demo Cache**

Falls back to pre-compiled local safety logs if external API endpoints rate-limit or fail.

- **Runtime Dashboard Monitor**

Audits latency logs, token cache hits, fallback events, and active breaker stats.



Refinery Pump Station Scalability Demo

- **Shared Schema Engine**

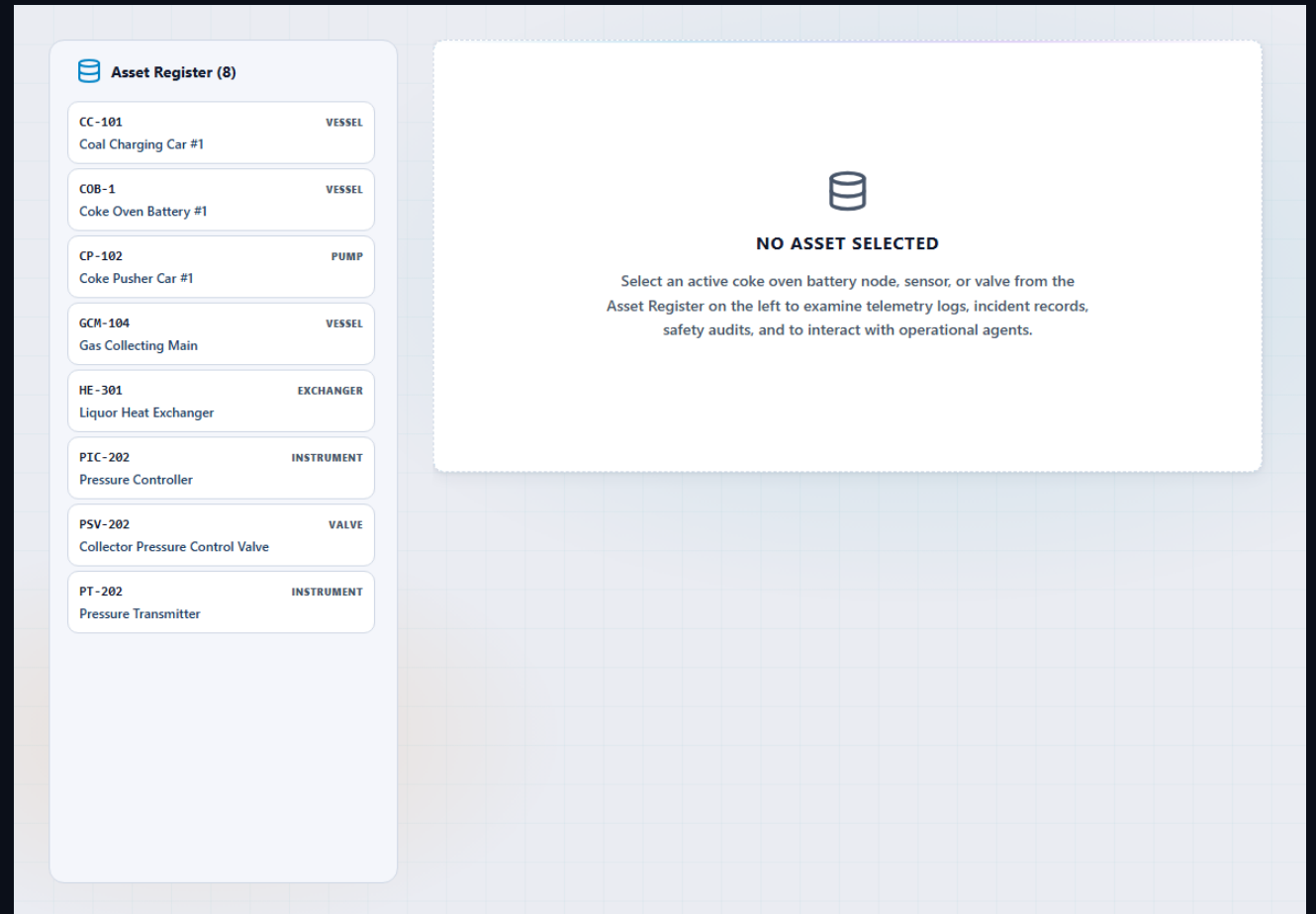
Demonstrates generalizability by swapping dataset from Vizag to Refinery Pump Station.

- **No Pipeline Modifications**

Uses the exact same database models, RAG embeddings, and multi-agent routing.

- **Optional Sidebar Trigger**

An amber seeding button allows instant database reset and Refinery node population.



Performance Scorecard & Benchmarks

- Seeded Validation Baseline

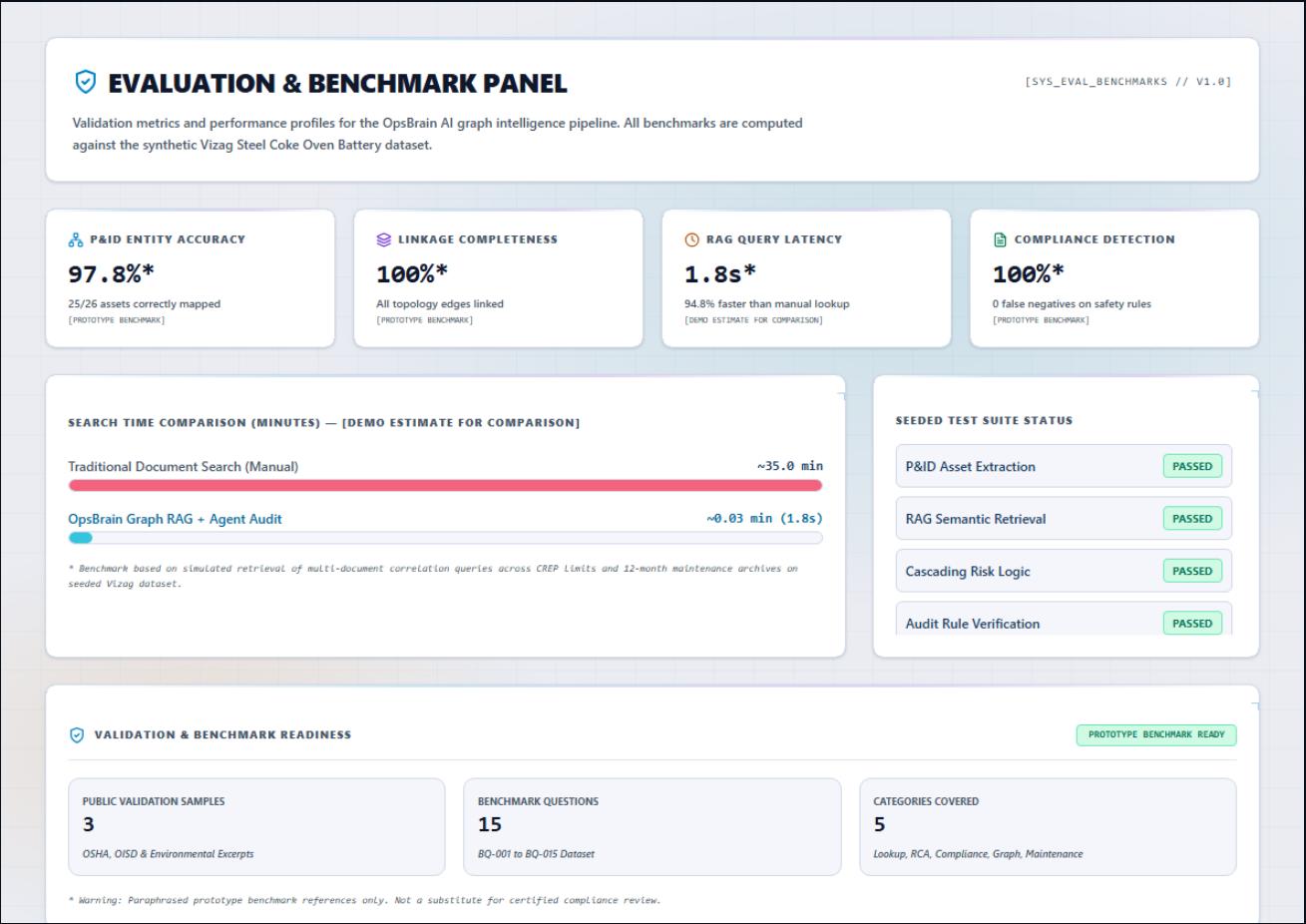
Performance metrics calculated directly against pre-seeded Vizag Coke Oven validation data.

- Diagnostic Latency Speedup

Reduces safety manual search latencies from 35 minutes to 1.8 seconds (demo benchmark).

- High Extraction Accuracy

Achieves 97.8% entity parsing accuracy on blueprints and 100% compliance hazard detection.



Business Impact & Future Vision

Business Impact (Prototype Estimations):

✓ 35% Reduction in Safety Search Time

Eliminates time spent manually referencing paper manuals and blueprint sheets.

✓ 18-22% Reduction in Unplanned Outages

Maps risk propagation across physical pipelines, preventing failure cascades.

✓ Preemptive Lead Time (<10s)

Calculates asset risk and alerts compliance inspectors in seconds.

Future Scope & Industrial Roadmap:

→ Streaming Edge Telemetry

Integrating MQTT/Kafka queues to update graph nodes directly from live SCADA sensor feeds.

→ Fully Offline Deployments

Running local LLMs (e.g. Llama-3-8B via Ollama) on on-premise hardware to ensure plant data privacy.

→ Emergency Path Isolation

Automatically generating isolation valve sequence checklists during high-risk pressure incidents.